Design Document: Outward Forex Remittance Engine (PS-016)

Team ID: T011
Problem Statement: PS-016 · Outward Forex Remittance Engine with LRS Limits, TCS and Rate Spreads
Domain: BFSI — Forex

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1. System Architecture & Pipeline Precedence (BR-11)

The engine enforces strict precedence for processing outward remittance requests:

2. Business Rules Specification

BR-01: LRS Limit & Residency

- **Cap:** USD 250,000 per financial year (April 1 – March 31) across all purposes per resident individual.
- Non-resident individuals are rejected outright.
- Non-USD transactions are converted to USD equivalent at the day's interbank rate to verify the cap.

BR-02: Purpose Codes & Permitted Activity

- Permitted Purpose Codes:
  - `S0305`: Education
  - `S1301`: Family Maintenance / Gift
  - `S0023`: Travel
  - `S0001`: Investment
  - `S1302`: Medical Treatment
- `S0001` (Investment) requires `FULL` customer KYC and a minimum amount of USD 1,000.

BR-03: Exchange Rates & Spreads

- Customer Rate = Interbank Rate×(1+Spread)

- Base Spreads:
  - USD, EUR, GBP: `0.50%` (`0.0050`)
  - AED, SGD: `0.75%` (`0.0075`)
  - All other currencies: `1.25%` (`0.0125`)
- High-volume discount: Spread is reduced by `0.10` percentage points (`-0.0010`) for remittances ≥USD 25,000.
- Quoted to 4 decimal places (`ROUND_HALF_UP`). Gross INR rounded to paisa (2 decimals).

BR-04: Cumulative Financial-Year TCS (Tax Collected at Source)

- TCS applies exclusively on the remittance INR amount (excluding bank charges/GST).
- Exemption limit: Cumulative remittances up to ₹10,00,000 in the FY attract `0%` TCS.
- Excess above ₹10,00,000:
  - Education funded by loan (`edu_loan_flag = True`): `0.5%`
  - Education or Medical without loan: `5.0%`
  - All other purposes (gifts, travel, investments): `20.0%`

BR-05: Bank Charges & GST Slabs

- Flat Service Charge: ₹500.00 + 18% GST (₹90.00) = ₹590.00.
- GST on Currency Conversion Slabs:
  - Gross ≤₹1,00,000: 0.18% of value (minimum ₹45.00).
  - ₹1,00,000<Gross≤₹10,00,000: ₹180.00 + 0.09% of excess over ₹1,00,000.
  - Gross >₹10,00,000: ₹990.00 + 0.018% of excess over ₹10,00,000 (maximum cap ₹10,800.00).

BR-06 & BR-07: Value Date & Cut-Off Logic

- Cut-off time: `14:30:00` on Indian business days.
- Requests before 14:30 receive same-day base T; requests on or after 14:30 roll to the next Indian business day.
- USD transfers >USD 10,000 carry an additional +1 business day for correspondent banking (BR-07).
- Value date rolls forward past any destination-country weekend or public holiday.

BR-08: Compliance HOLD

- Single remittance >USD 50,000 or aggregate FY remittance >USD 150,000 triggers `HOLD`.
- Clears with `APPROVE <txn_id>` or auto-rejects after 2 business days.

BR-09: Cancellations

- Requests cancelled before 14:30 on the booking day: Free cancellation.
- Requests cancelled after 14:30 on booking day: 1% penalty on Gross INR.
- Cancellations are rejected if requested after the value date.

BR-10: Rate Recheck (RATE_RECHECK)

- If market interbank rate increases >1.00 against the bank before the value date, the trade is re-priced and the customer is debited the differential. Moves in the customer's favour are not credited back.